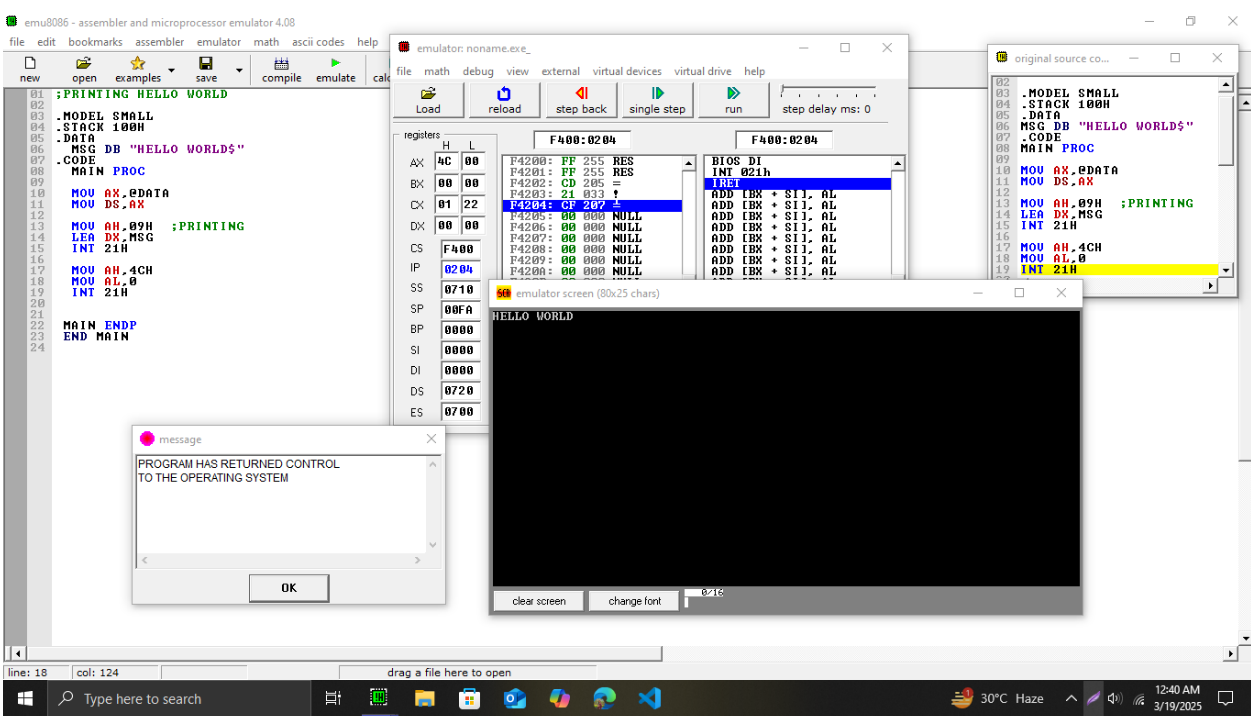
LAB-1

1) ;PRINTING HELLO WORLD

```
.MODEL SMALL
 .STACK 100H
 .DATA
  MSG DB "HELLO WORLD$"
 .CODE
   MAIN PROC
    MOV AX,@DATA
    MOV DS,AX
    MOV AH,09H     ;PRINT
    LEA DX,MSG
    INT 21H
     MOV AH,4CH        ;ENDING
    MOV AL,0
    INT 21H
  MAIN ENDP
 END MAIN
```

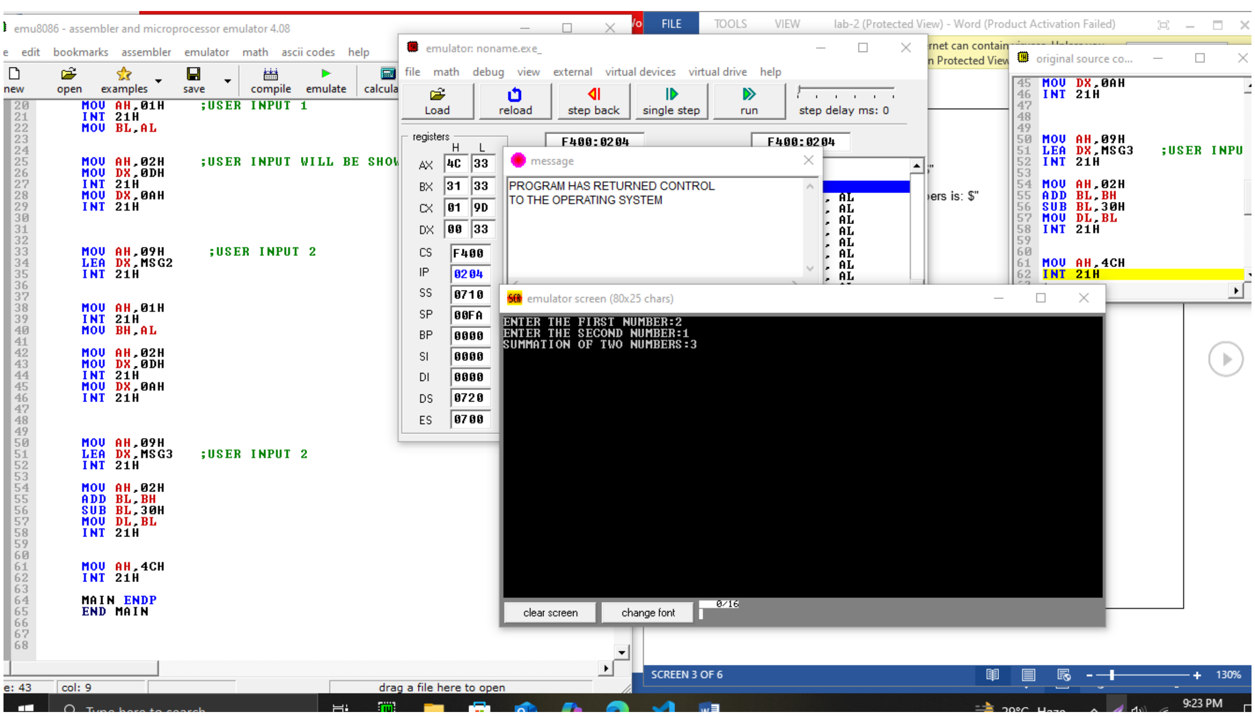


LAB-2

1) ;ADDITION OF TWO NUMBERS TAKING INPUT FROM USER

```
.MODEL SMALL
.STACK 100H
.DATA
  MSG1 DB "ENTER 1ST NUMBER:$"
  MSG2 DB "ENTER 2ND NUMBER:$"
  MSG3 DB "SUMMATION OF NUMBERS:$"
.CODE
  MAIN PROC
   MOV AX,@DATA
   MOV DS,AX
   MOV AH,09H     ;PRINT
   LEA DX,MSG1
   INT 21H
   MOV AH,01H     ;TAKING INPUT FROM USER
   INT 21H
   MOV AH,02H   ;DISPLAYING USER INPUT AND SHIFT TO A NEW LINE
   MOV DX,0DH
   INT 21H
   MOV DX,0AH
   INT 21H
    MOV AH,09H
   LEA DX,MSG2
   INT 21H
MOV AH,01H
   INT 21H
   MOV AH,01H
   INT 21H
   MOV BH,AL
```

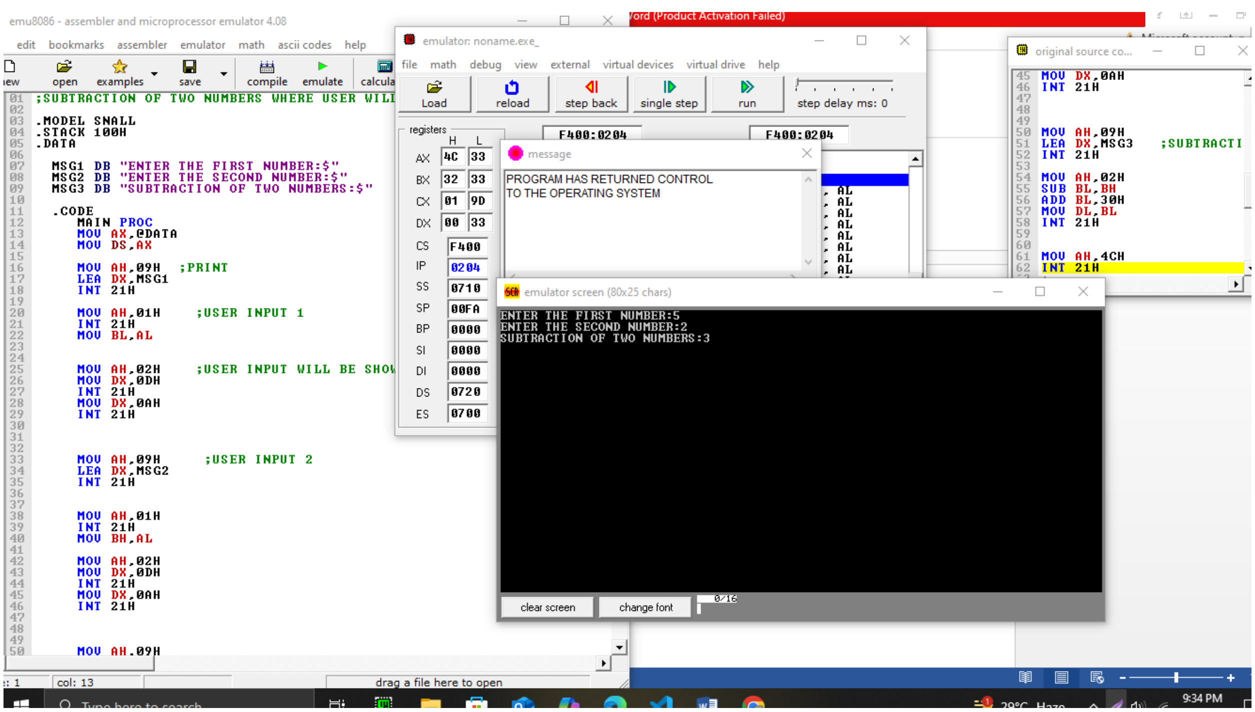
```
    MOV AH,02H
    MOV AX,0DH
    INT 21H
    MOV DX,0AH
    INT 21H
    MOV AH,09H
    LEA DX,MSG3
    INT 21H
    MOV AH,02H
    ADD BL,BH
    SUB BL,30H
    MOV DL,BL
    INT 21H
    MOV AH,4CH        ;ENDING
    INT 21H
MAIN ENDP
END MAIN
```



2);SUBTRACTION OF TWO NUMBER USING USER INPUT

```
 .MODEL SMALL
.STACK 100H
.DATA
   MSG1 DB "ENTER 1ST NUMBER:$"
   MSG2 DB "ENTER 2ND NUMBER:$"
   MSG3 DB "SUBTRACTION RESULT:$"
.CODE
   MAIN PROC
    MOV AX,@DATA
    MOV DS,AX
    MOV AH,09H     ; PRINT
    LEA DX,MSG1
    INT 21H
    MOV AH,01H     ; TAKING INPUT FROM USER
    INT 21H
    MOV BH,AL      ; STORE FIRST NUMBER
   MOV AH,02H     ; DISPLAYING USER INPUT AND SHIFT TO A NEW LINE
    MOV DX,0DH
    INT 21H
    MOV DX,0AH
    INT 21H
     MOV AH,09H
    LEA DX,MSG2
    INT 21H
  MOV AH,01H
    INT 21H
    MOV BL,AL
      MOV AH,02H
    MOV DX,0DH
    INT 21H
```

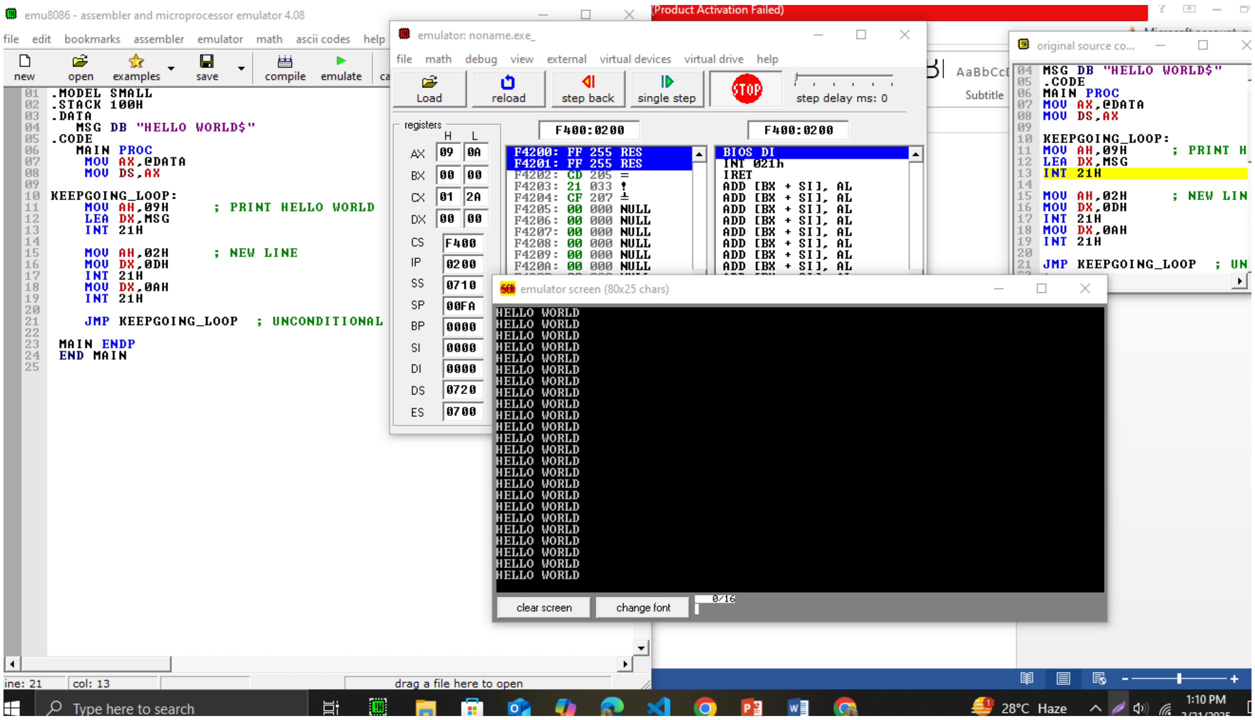
```
MOV DX,0AH
INT 21H
MOV AH,09H
LEA DX,MSG3
INT 21H
MOV AH,02H
SUB BH,BL     ; PERFORMING SUBTRACTION
ADD BH,30H
MOV DL,BH
INT 21H
MOV AH,4CH       ; ENDING
INT 21H
 MAIN ENDP
END MAIN
```



LAB-3

1);PRINTING HELLO WORLD INFINITE TIMES

```
.MODEL SMALL
.STACK 100H
.DATA
 MSG DB "HELLO WORLD$"
.CODE
 MAIN PROC
  MOV AX,@DATA
  MOV DS,AX
 KEEPGOING_LOOP:
  MOV AH,09H     ; PRINT HELLO WORLD
  LEA DX,MSG
  INT 21H
  MOV AH,02H     ; NEW LINE
  MOV DX,0DH
  INT 21H
  MOV DX,0AH
  INT 21H
  JMP KEEPGOING_LOOP  ; UNCONDITIONAL JUMP BACK TO KEEPGOING_LOOP
   MAIN ENDP
 END MAIN
```



2) .MODEL SMALL

.STACK 100H

.DATA

MSG1 DB "HELLO WORLD$"

MSG2 DB "bYE BYE WORLD.$"

.CODE

MAIN PROC

MOV AX,@DATA

MOV DS,AX

CHECK_CONDITION:

CMP BX,5H

JG STOP

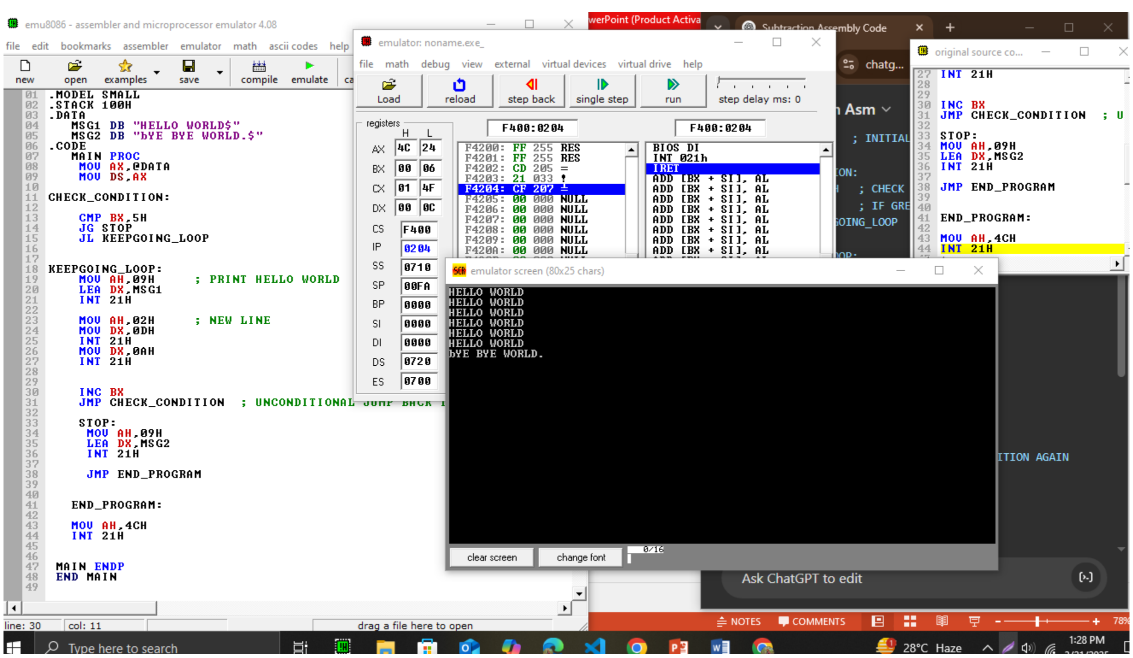
JL KEEPGOING_LOOP

KEEPGOING_LOOP:

MOV AH,09H ; PRINT HELLO WORLD

LEA DX,MSG1

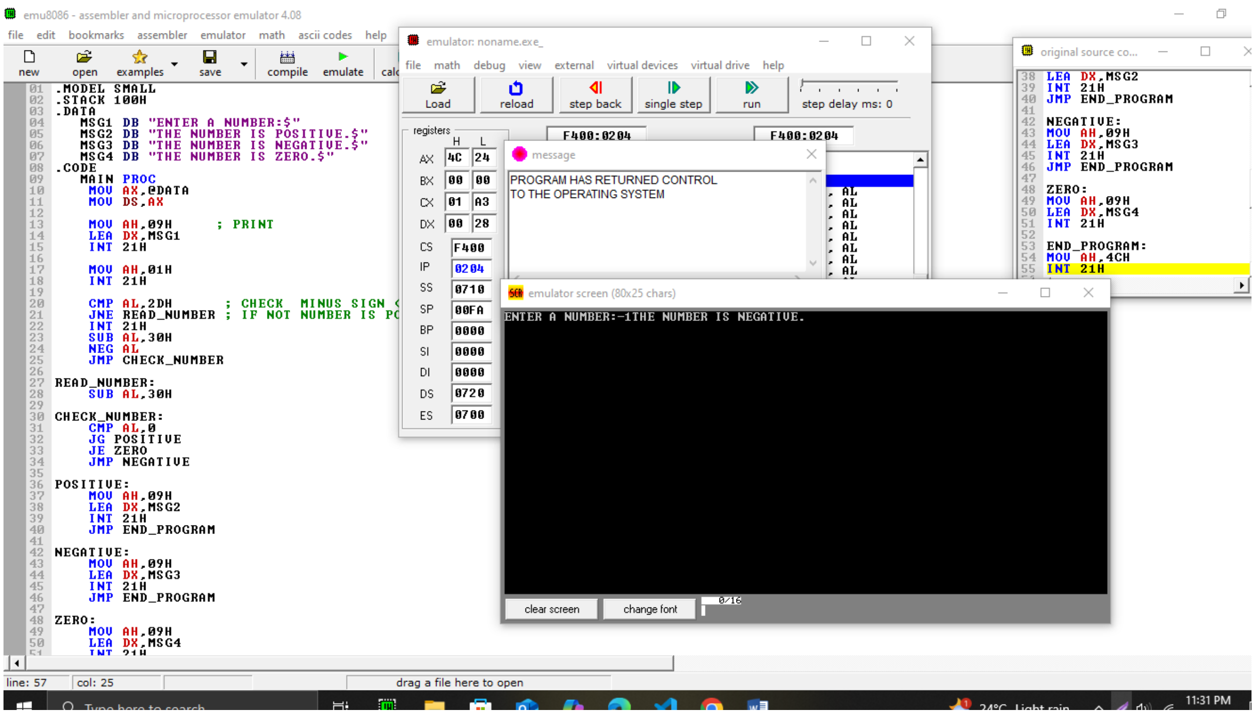
```
        INT 21H
 MOV AH,02H
 MOV DX,0DH
        INT 21H
        MOV DX,0AH
        INT 21H
INC BX
JMP CHECK_CONDITION  ; UNCONDITIONAL JUMP BACK TO KEEPGOING_LOOP
STOP:
        MOV AH,09H
        LEA DX,MSG2
        INT 21H
JMP END_PROGRAM
END_PROGRAM:
MOV AH,4CH
INT 21H
MAIN ENDP
END MAIN
```

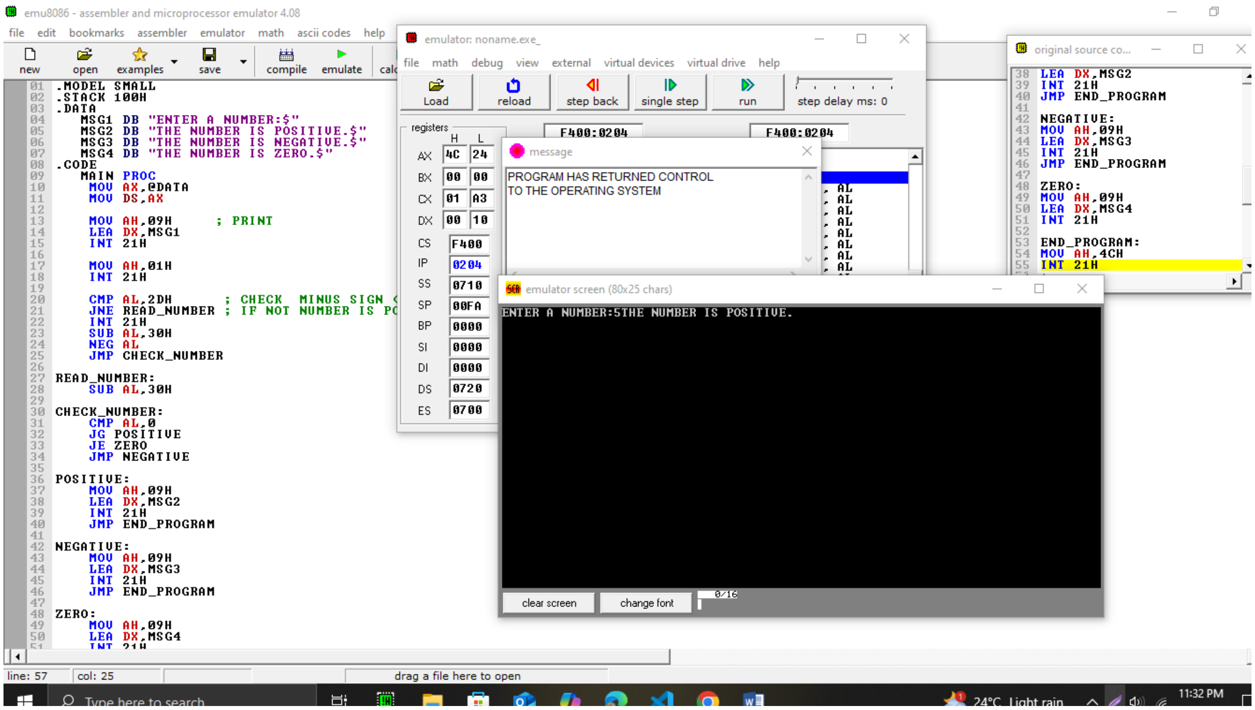


3);CHECK NUMBER POSITIVE/NEGATIVE/ZERO

```
 .MODEL SMALL
.STACK 100H
.DATA
   MSG1 DB "ENTER A NUMBER:$"
   MSG2 DB "THE NUMBER IS POSITIVE.$"
   MSG3 DB "THE NUMBER IS NEGATIVE.$"
   MSG4 DB "THE NUMBER IS ZERO.$"
.CODE
   MAIN PROC
    MOV AX,@DATA
    MOV DS,AX
MOV AH,09H     ; PRINT
    LEA DX,MSG1
    INT 21H
      MOV AH,01H
    INT 21H
     CMP AL,2DH      ; CHECK  MINUS
    JNE READ_NUMBER ; IF NOT NUMBER IS POSITIVE
    INT 21H
    SUB AL,30H
    NEG AL
    JMP CHECK_NUMBER
READ_NUMBER:
    SUB AL,30H
CHECK_NUMBER:
    CMP AL,0
    JG POSITIVE
    JE ZERO
    JMP NEGATIVE
POSITIVE:
```

```
   MOV AH,09H
   LEA DX,MSG2
   INT 21H
   JMP END_PROGRAM
NEGATIVE:
   MOV AH,09H
   LEA DX,MSG3
   INT 21H
   JMP END_PROGRAM
ZERO:
   MOV AH,09H
   LEA DX,MSG4
   INT 21H
END_PROGRAM:
   MOV AH,4CH
   INT 21H
    MAIN ENDP
 END MAIN
```

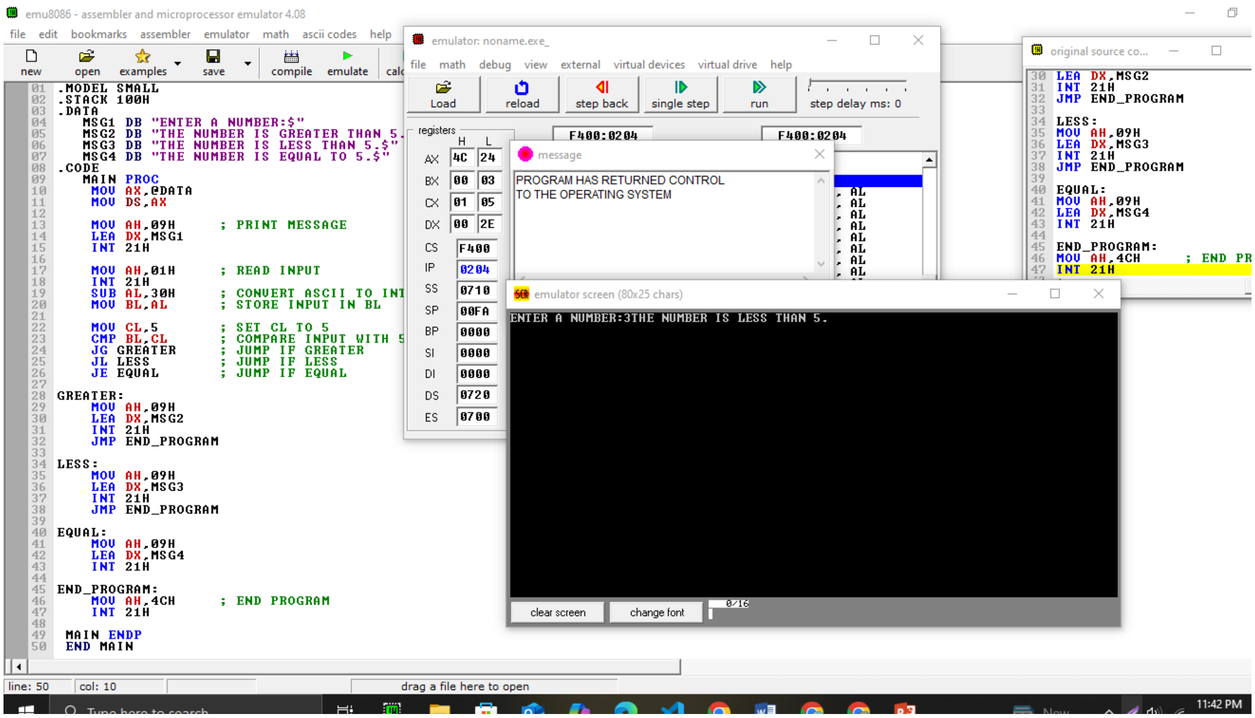




4);CHECKING NUMBER COMPARING WITH 5

```
 .MODEL SMALL
.STACK 100H
.DATA
   MSG1 DB "ENTER A NUMBER:$"
   MSG2 DB "THE NUMBER IS GREATER THAN 5.$"
   MSG3 DB "THE NUMBER IS LESS THAN 5.$"
   MSG4 DB "THE NUMBER IS EQUAL TO 5.$"
.CODE
   MAIN PROC
    MOV AX,@DATA
    MOV DS,AX
MOV AH,09H     ; PRINT MESSAGE
    LEA DX,MSG1
    INT 21H
     MOV AH,01H     ; READ INPUT
    INT 21H
```

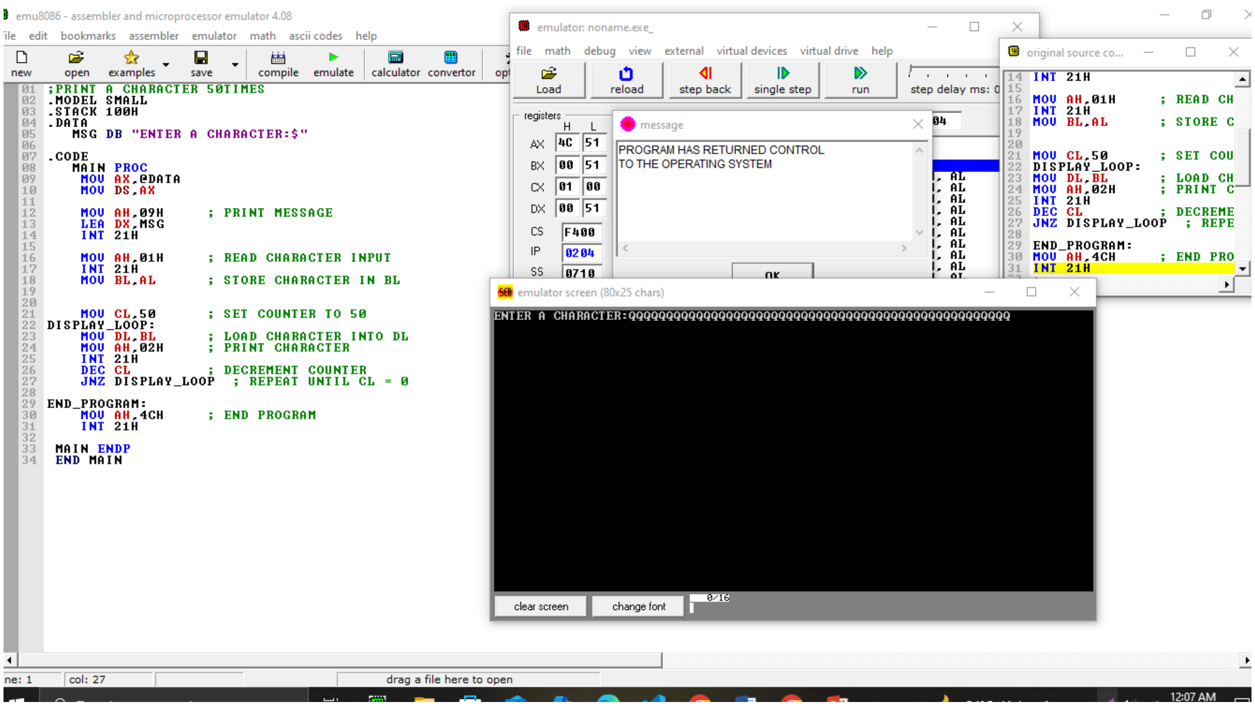
```
   SUB AL,30H     ; CONVERT ASCII TO INTEGER
   MOV BL,AL      ; STORE INPUT IN BL
 MOV CL,5       ; SET CL TO 5
   CMP BL,CL      ; COMPARE INPUT WITH 5
   JG GREATER     ; JUMP IF GREATER
   JL LESS        ; JUMP IF LESS
   JE EQUAL       ; JUMP IF EQUAL
GREATER:
   MOV AH,09H
   LEA DX,MSG2
   INT 21H
   JMP END_PROGRAM
LESS:
   MOV AH,09H
   LEA DX,MSG3
   INT 21H
   JMP END_PROGRAM
EQUAL:
   MOV AH,09H
   LEA DX,MSG4
   INT 21H
END_PROGRAM:
   MOV AH,4CH     ; END PROGRAM
   INT 21H
   MAIN ENDP
 END MAIN
```



6) ;PRINT A CHARACTER 50TIMES

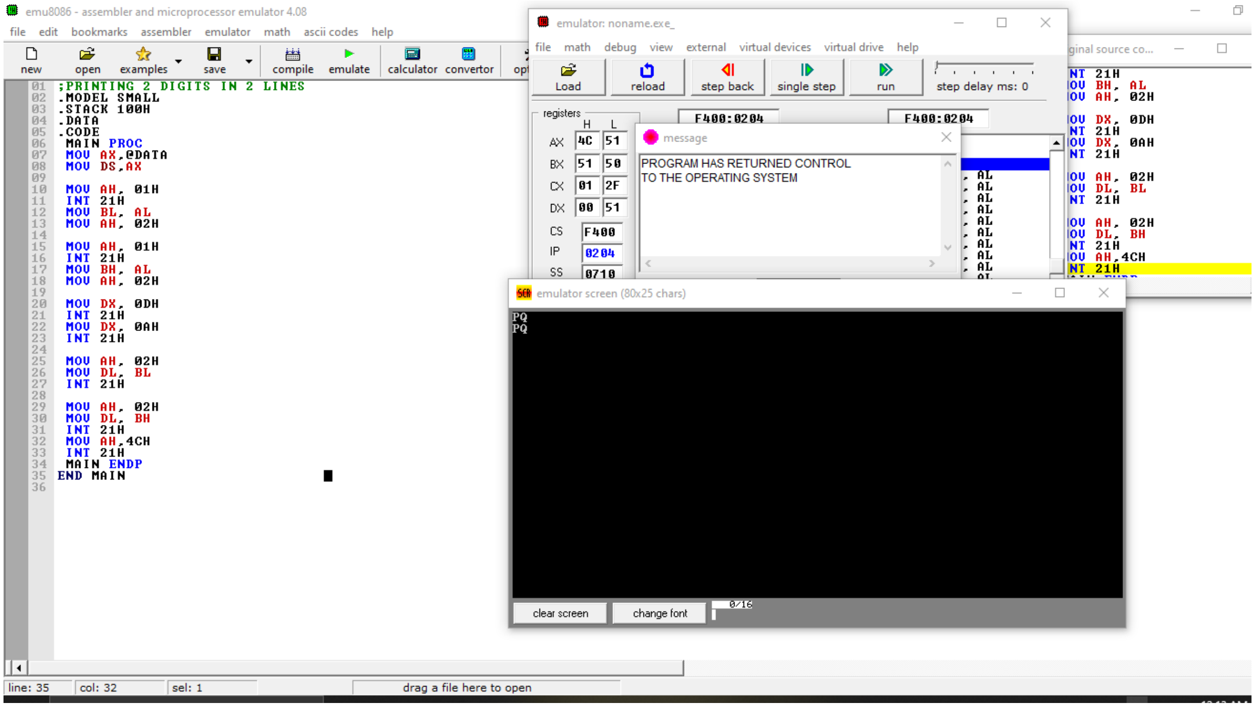
```
.MODEL SMALL
.STACK 100H
.DATA
  MSG DB "ENTER A CHARACTER:$"
 .CODE
  MAIN PROC
   MOV AX,@DATA
   MOV DS,AX
MOV AH,09H     ; PRINT MESSAGE
   LEA DX,MSG
   INT 21H
   MOV AH,01H     ; READ CHARACTER INPUT
   INT 21H
   MOV BL,AL      ; STORE CHARACTER IN BL
 MOV CL,50      ; SET COUNTER TO 50
```

```
DISPLAY_LOOP:
    MOV DL,BL      ; LOAD CHARACTER INTO DL
    MOV AH,02H     ; PRINT CHARACTER
    INT 21H
    DEC CL         ; DECREMENT COUNTER
    JNZ DISPLAY_LOOP  ; REPEAT UNTIL CL = 0
END_PROGRAM:
    MOV AH,4CH     ; END PROGRAM
    INT 21H
    MAIN ENDP
 END MAIN
```



7);PRINTING 2 DIGITS IN 2 LINES

```
 .MODEL SMALL
.STACK 100H
.DATA
.CODE
 MAIN PROC
 MOV AX,@DATA
 MOV DS,AX
 MOV AH, 01H
 INT 21H
 MOV BL, AL
 MOV AH, 02H
 MOV AH, 01H
 INT 21H
 MOV BH, AL
 MOV AH, 02H
 MOV DX, 0DH
 INT 21H
 MOV DX, 0AH
 INT 21H
 MOV AH, 02H
 MOV DL, BL
 INT 21H
 MOV AH, 02H
 MOV DL, BH
 INT 21H
 MOV AH,4CH
 INT 21H
 MAIN ENDP
END MAIN
```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;PRINTING 2 DIGITS IN 2 LINES
.MODEL SMALL
.STACK 100H
.DATA
.CODE
MAIN PROC
MOV AX,@DATA
MOV DS,AX
MOV AH, 01H
INT 21H
MOV BL, AL
MOV AH, 02H
MOV AH, 01H
INT 21H
MOV BH, AL
MOV AH, 02H
MOV DX, 0DH
INT 21H
MOV DX, 0AH
INT 21H
MOV AH, 02H
MOV DL, BL
INT 21H
MOV AH, 02H
MOV DL, BH
INT 21H
MOV AH,4CH
INT 21H
MAIN ENDP
END MAIN
emulator: noname.exe_
file math debug view external virtual devices virtual drive help
Load reload step back single step run step delay ms: 0
registers
AX 4C 51
BX 51 50
CX 01 2F
DX 00 51
CS F400
IP 0204
SS 0710
F400:0204
message
PROGRAM HAS RETURNED CONTROL
TO THE OPERATING SYSTEM
emulator screen (80x25 chars)
PQ
PQ
clear screen
change font
line: 35
col: 32
sel: 1
drag a file here to open